

YASH ENTERPRISES

Platform Architecture, Technology Stack & Database Schema Documentation

Platform Name: Yash Enterprises Security & Biometric Solutions Portal
Live Domain: bestcctvservice.com | **Support Phone:** +91 9308907319
Environment: Production / Enterprise Grade Web Application
Document Purpose: Complete technical specification of technology stack, system architecture, and Relational Database Management System (MySQL) schema design.

1. Technology Stack Architecture

The platform is built using modern full-stack web standards ensuring high performance, SEO optimization, and secure operations.

Layer / Domain	Technologies & Frameworks	Role & Key Highlights
Frontend Framework	Next.js 14 (App Router), React 18, TypeScript	Server-Side Rendering (SSR), Static Generation (SSG), dynamic SEO tags, strict typing.
Styling & UI	Tailwind CSS, Lucide React, Responsive Grid	Custom enterprise UI theme, mobile-first responsive layout, accessible modal portals.
Backend API Layer	Next.js Route Handlers (Node.js runtime)	RESTful API endpoints for leads, tickets, AMC contracts, products, and admin auth.
Database Engine	MySQL 8.0 Community / Server (Port 3306)	ACID compliant relational database, foreign key relations, auto-indexing on primary keys.
ORM & Query Layer	Prisma ORM v5	Schema-first migrations, type-safe database queries, connection pool management.
Security & Auth	JSON Web Tokens (JWT), Bcrypt password hashing	Role-Based Access Control (SUPERADMIN / ADMIN / TECHNICIAN), secure HTTP cookies.
Media & Branding	Local static assets + Scalable SVG vector icons	High-resolution Yash Enterprises logo, official WhatsApp vector emblem, product images.

2. Database Specifications

Database Engine: MySQL Relational Database (RDBMS)
Database Name: bt_security_db
Default Port: 3306 | **Encoding:** utf8mb4 | **Collation:** utf8mb4_unicode_ci
Connection String Pattern: mysql://root:password@localhost:3306/bt_security_db
Schema Definition File: prisma/schema.prisma

3. Complete Schema & Table Structures

The database contains **5 core production tables** managing staff credentials, product catalog, customer inquiries, service tickets, and AMC maintenance contracts.

1. admin_users

Purpose: Stores administrative login credentials, role permissions (SUPERADMIN, ADMIN, TECHNICIAN), and timestamps.

Column Name	Data Type	Key / Constraint	Description & Business Rule
id	VARCHAR(191)	PRIMARY KEY	UUID string generated via cuid()
name	VARCHAR(191)	NOT NULL	Full display name of the staff / administrator
email	VARCHAR(191)	UNIQUE NOT NULL	Unique email address used for portal login
passwordHash	VARCHAR(191)	NOT NULL	Bcrypt hashed password (salted and encrypted)
role	ENUM/VARCHAR(50)	DEFAULT "ADMIN"	Role: SUPERADMIN, ADMIN, or TECHNICIAN
createdAt	DATETIME(3)	DEFAULT NOW()	Timestamp when account was created
updatedAt	DATETIME(3)	ON UPDATE NOW()	Automatic timestamp of last profile / role update

2. products

Purpose: Inventory & catalog for CCTV cameras (HD Analog, IP, PTZ, Dome, Bullet), Biometric machines, and Accessories.

Column Name	Data Type	Key / Constraint	Description & Business Rule
id	VARCHAR(191)	PRIMARY KEY	Unique product ID
name	VARCHAR(191)	NOT NULL	Item display name (e.g. 2MP Full HD Dome Camera)
slug	VARCHAR(191)	UNIQUE NOT NULL	URL-friendly SEO slug (e.g. 2mp-dome-camera)
category	VARCHAR(191)	NOT NULL	Category: CCTV, BIOMETRIC, ACCESS_CONTROL, ACCESSORY
brand	VARCHAR(191)	NOT NULL	Manufacturer brand: CP Plus, Hikvision, Dahua, eSSL, Realtime
modelNumber	VARCHAR(191)	NULLABLE	Manufacturer hardware model / serial identifier
price	DECIMAL(10,2)	DEFAULT 0.00	Base retail/display price in INR
description	TEXT	NOT NULL	Detailed specifications and technical datasheet text
features	TEXT / JSON	NULLABLE	Bullet points / feature highlights in JSON format
imageUrl	VARCHAR(500)	NULLABLE	URL to the product image
inStock	BOOLEAN	DEFAULT TRUE	Availability indicator for customer orders
featured	BOOLEAN	DEFAULT FALSE	Flag to showcase product on homepage

3. leads

Purpose: Captures customer quotations, website inquiry forms, callback requests, and installation requirements.

Column Name	Data Type	Key / Constraint	Description & Business Rule
id	VARCHAR(191)	PRIMARY KEY	Unique lead ID
name	VARCHAR(191)	NOT NULL	Prospect / Customer full name
phone	VARCHAR(20)	NOT NULL	Primary mobile / WhatsApp contact number
email	VARCHAR(191)	NULLABLE	Customer email address for sending quotations
serviceType	VARCHAR(100)	NOT NULL	CCTV Installation, Biometric Setup, AMC, Repair, Custom
message	TEXT	NULLABLE	Customer notes, camera count, or location details
status	VARCHAR(50)	DEFAULT "NEW"	Lead status: NEW, CONTACTED, QUOTED, WON, LOST
createdAt	DATETIME(3)	DEFAULT NOW()	Timestamp when enquiry was submitted

4. service_tickets

Purpose: Doorstep repair & technician dispatch management. Tracks ticket number, customer address, issue description, and technician status.

Column Name	Data Type	Key / Constraint	Description & Business Rule
id	VARCHAR(191)	PRIMARY KEY	Unique ticket record ID
ticketNumber	VARCHAR(50)	UNIQUE NOT NULL	Public tracking ID formatted as TIC-YYYY-XXXX
customerName	VARCHAR(191)	NOT NULL	Full name of customer needing service
phone	VARCHAR(20)	NOT NULL	Customer contact number for technician coordination
address	TEXT	NOT NULL	Doorstep physical installation / repair address
city	VARCHAR(100)	DEFAULT "Patna"	Service city / locality
serviceType	VARCHAR(100)	NOT NULL	CCTV Offline, DVR Beeping, HDD Failure, Biometric Sync
issueDescription	TEXT	NOT NULL	Exact fault details described by customer
status	VARCHAR(50)	DEFAULT "OPEN"	OPEN, ASSIGNED, IN_PROGRESS, RESOLVED, CANCELLED
technicianName	VARCHAR(191)	NULLABLE	Assigned field technician name
technicianPhone	VARCHAR(20)	NULLABLE	Field technician contact number
scheduledDate	DATETIME(3)	NULLABLE	Scheduled visit date and time window
resolutionNotes	TEXT	NULLABLE	Technician visit report and parts replaced
createdAt	DATETIME(3)	DEFAULT NOW()	Ticket creation timestamp

5. amc_contracts

Purpose: Annual Maintenance Contracts (AMC) for corporate offices, retail stores, residential apartments, and warehouses.

Column Name	Data Type	Key / Constraint	Description & Business Rule
id	VARCHAR(191)	PRIMARY KEY	Unique contract ID
contractNumber	VARCHAR(50)	UNIQUE NOT NULL	AMC agreement number formatted as AMC-YYYY-XXXX
customerName	VARCHAR(191)	NOT NULL	Signatory / Contact person name
companyName	VARCHAR(191)	NULLABLE	Business / Organization / Apartment society name
phone	VARCHAR(20)	NOT NULL	Primary billing & support phone number
email	VARCHAR(191)	NULLABLE	Official email for renewal notices and billing invoices
address	TEXT	NOT NULL	Contract site location / premises address
planType	VARCHAR(50)	NOT NULL	Comprehensive, Non-Comprehensive, Quarterly Visit Plan
totalCameras	INT	DEFAULT 0	Number of CCTV cameras covered under this AMC
totalBiometrics	INT	DEFAULT 0	Number of Biometric attendance machines covered
amount	DECIMAL(10,2)	NOT NULL	Total annual contract value in INR
startDate	DATETIME(3)	NOT NULL	Contract effective inception date
endDate	DATETIME(3)	NOT NULL	Contract expiration / renewal due date
status	VARCHAR(50)	DEFAULT "ACTIVE"	ACTIVE, EXPIRING_SOON, EXPIRED, TERMINATED
notes	TEXT	NULLABLE	Custom clauses, SLA response times (e.g. 4-hour SLA)

4. Maintenance, Backup & Operational Commands

MySQL Database Backup Command (CLI):
mysqldump -u root -p bt_security_db > backup_bt_security_db.sql

Prisma Schema Push & Sync:
npx prisma db push (Sync schema directly to MySQL database)
npx prisma studio (Open GUI database viewer on port 5555)

Application Deployment & Run:
npm run build (Production compile)
npm run start (Launches production web server on port 3000)